

ADITYA THAKARE

Data Scientist | Generative AI | MLOps

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SUMMARY

Data Scientist with 2+ years of experience designing and deploying production-grade Generative AI systems — including RAG pipelines, multi-agent LangGraph orchestration, and end-to-end MLOps workflows. Proficient in LangGraph, LangChain, FastAPI, Docker, and MLflow. Built and deployed three fully operational LLM-powered systems (legal AI, agentic research, predictive maintenance) with live production endpoints. Delivered technical training to 2,000+ professionals across 10+ industry engagements. Research background in deep learning-based ancient script recognition and applied NLP/ML.

SKILLS

Languages	Python, SQL
Generative AI	RAG · Agentic AI · LLM · LangGraph · LangChain · Gemini API · OpenAI API · Prompt Engineering · Vector Databases
Vector Stores	ChromaDB · Pinecone · HuggingFace · Gemini Embeddings · Semantic Chunking
ML / Deep Learning	Scikit-learn · XGBoost · PyTorch · TensorFlow · Keras · CNN · NLP · Transfer Learning
Backend & APIs	FastAPI · REST APIs · Streamlit · Pydantic
MLOps & DevOps	Docker · Nginx · GitHub Actions · CI/CD · MLflow · DVC · SystemD · SSL/HTTPS
Data & Visualization	Pandas · NumPy · Matplotlib · Seaborn · Power BI · Tableau

PROFESSIONAL EXPERIENCE

Data Scientist | MITU Skillologies

2023 – Present | Pune, India

Assisted in deep learning research on **Brahmi script recognition** — conducted literature review across OCR and CNN methodologies; supported baseline implementation across four architectures (Custom Deep CNN, InceptionNet, ResNet-50, XceptionNet) on a **7,880-image, 56-class dataset**; ResNet-50 selected as optimal with **highest benchmark accuracy** across all evaluated models.

Contributed research and implementation support to applied ML projects including **student progression prediction** (multi-source data integration, feature analysis) and **crop disease classification** — comparative experimentation across multiple model architectures.

Designed and built **end-to-end AI systems** demonstrating LangChain RAG pipelines, neural network architectures, and MLOps workflows — serving as **primary technical SME across 10+ industry engagements reaching 2,000+ professionals** including corporate workshops and faculty development programs.

Architected and delivered a **140-session full-stack AI curriculum** (Python through Generative AI and LangGraph) and independently led a 3-hour technical session on "Next-Gen AI: Neural Networks & Applications" for mixed technical and non-technical audiences.

PROJECTS

Conversational Legal Agent (RAG)

 Live Demo |  Source Code

Addresses the lack of affordable legal information access for individuals who cannot afford legal consultation.

Engineered and deployed an end-to-end conversational RAG agent specialized in Indian law. Built a complete data pipeline — web scraping, preprocessing, and semantic chunking of legal datasets. Chose **ChromaDB** over FAISS for persistent storage, enabling stateful retrieval without re-embedding. Used **Gemini text-embedding** for domain-appropriate semantic representation. Deployed via Streamlit + Nginx on DigitalOcean with SSL and systemd process management.

Tech: RAG · LangChain · ChromaDB · Gemini API · Streamlit · Nginx · DigitalOcean

Agentic Research System (Multi-Agent AI)

[🔗 Live Demo](#) | [⚙️ Source Code](#)

Solves shallow, single-pass AI research outputs via a self-refining multi-agent system that iterates until quality thresholds are met.

Built a production multi-agent system using **LangGraph** — chosen over LangChain's AgentExecutor for native cyclic graph support, persistent typed state, and deterministic control flow. Three specialized agents: Researcher (Tavily + Gemini synthesis), Writer (structured drafting), and Critic (quality scoring with revision trigger). **TypedDict state schema** with conditional edges loops back through Writer only when Critic score falls below threshold — preventing infinite loops and premature termination.

Tech: Agentic AI · LangGraph · LangChain · Gemini API · Tavily Search · Streamlit · Docker

Industrial Equipment Maintenance (End-to-End MLOps)

[🔗 Live Demo](#) | [⚙️ Source Code](#)

Targets the high cost of unplanned manufacturing downtime via sensor-driven predictive maintenance.

Developed a predictive maintenance classifier using **XGBoost** — selected over neural alternatives for tabular sensor data in low-data regimes. Fully decoupled microservices architecture: **FastAPI** for model inference, **Streamlit** for operator visualization, **MLflow** for experiment tracking, and **DVC** for data lineage. Deployed on DigitalOcean with Nginx reverse proxy, SSL termination, and systemd-based process management.

Tech: XGBoost · FastAPI · MLflow · DVC · Streamlit · Docker · Nginx · SystemD · DigitalOcean

CEO Financial Dashboard (Agentic AI Backend)

Production / Client Project

Production-grade financial analytics dashboard with natural-language query capabilities for C-suite executives.

Architected an agentic AI backend using **LangGraph** orchestration with **FastAPI** serving layer. Integrated **DuckDB/Parquet** for high-performance analytical queries and **MongoDB** for metadata storage. Implemented structured outputs via **Pydantic** models with GPT-4o-mini for reliable, schema-validated natural language responses to financial queries.

Tech: LangGraph · FastAPI · DuckDB · Parquet · MongoDB · GPT-4o-mini · Pydantic

CERTIFICATIONS

Supervised Machine Learning: Regression and Classification — DeepLearning.AI, Stanford University	Nov 2024
Generative AI, LLM & RAG — GeeksforGeeks	Dec 2025
Prompt Engineering for ChatGPT — Vanderbilt University	Mar 2024
Generative AI for Everyone — Coursera	Nov 2023
Databases and SQL for Data Science with Python — Coursera	May 2023
Docker Training Course for the Absolute Beginner — KodeKloud	Mar 2023

EDUCATION

PG Diploma in Artificial Intelligence CDAC ACTS, Pune	2023
B.Tech – Civil Engineering (CGPA: 7.78) Savitribai Phule Pune University	2021